FUEL ANALYSIS REPORT

REPORT ID: S261083440	CUSTOMER: HUAYU SHIP MANAGEMENT LIMITED	SHIP(IMO NO): XINSHUN(9252199)	BUNKER PORT: UST – LUGA

CONFORMANCE

The fuel sample tested conforms to Table 1 of ISO 8217:2005 specifications for grade MGO - DMA LS

*Conformance to table 1 or table 2 of ISO 8217 does not imply conformance to other sections of ISO 8217 such as para 5.

BUNKER INFORMATION		BDN/CLIENT INFORMATION		SAMPLE DETAILS	
BUNKER DATE	21-Oct-2026	GRADE	DMA LS	SENT FROM	
BUNKER PORT	UST – LUGA	DENSITY@15°C kg/m³	829.0	SENT DATE	26-Oct-2026
SUPPLIER	BALTBUNKER	VISCOSITY@40°C cSt	2.707	REPORT DATE	15-Nov-2026
BARGE	REVANCHE	SULFUR %m/m	0.00054	TEST DATE	14-Nov-2026
SAMPLING POINT	MANIFOLD	WATER %v/v	-	RETENTION DATE	14-Dec-2026
SAMPLING METHOD	DRIP	QUANTITY (MT)	410.000	PO NUMBER	-

SEAL INFORMATION

LAB SEAL	MARPOL SEAL	SHIP SEAL	BARGE SEAL
D593358(Sealed)	D593355		

RECOMMENDED TEMPERATURES

A minimum viscosity has to be maintained at engine inlet and for practical purpose some engine makers recommend this to be not less than 2cSt. Based on tested viscosity, this oil should not be heated above 78°C temperature in order to meet the requirement.

CALCULATIONS OF QUANTITY SUPPLIED BASED ON LAB TESTS

	UNITS	BUNKER INFORMATION	TESTED RESULTS
Density @15°C	kg/m³	829.0	828.5
Water	%v/v	-	0.000
Quantity supplied as per ship	MT	410.000	
Quantity shortage due to density	MT	0.247	
Quantity shortage due to water	MT	-	
Total Quantity Shortage	MT	0.247	

*Above figures do not account for ISO 4259 criteria given in ISO 8217 Annex F. (This is not applicable for delivery made by mass flowmeter)

TEST DESCRIPTION	UNITS	RESULTS	ISO 8217 LIMITS	METHOD
Density @15°C	kg/m³	828.5	Max 890.0	ISO 12185
Viscosity @ 40°C	cSt	2.69	Min 1.5 Max 6.0	ISO 3104
Flash Point	°C	> 70.0	Min 60.0	ISO 3679
Pour Point	°C	0	Max(Summer) 0 Max(Winter) -6	ISO 3016
Micro Carbon Residue	% m/m	0.01	Max 0.30	ISO 10370
Ash	% m/m	0.003	Max 0.010	Calculated
Water	% v/v	< 0.01	-	ASTM D6304
Sulfur	% m/m	0.00056	Max 0.10	ISO 8754
Vanadium	mg/kg	< 1	-	VLS-LM8
Al+Si (Catfines)	mg/kg	2	-	VLS-LM8
Sodium	mg/kg	< 1	-	VLS-LM8
Aluminum	mg/kg	< 1	-	VLS-LM8
Silicon	mg/kg	< 1	-	VLS-LM8
Iron	mg/kg	< 1	-	VLS-LM8
Nickel	mg/kg	< 1	-	VLS-LM8
Calcium+	mg/kg	< 1	-	VLS-LM8
Phosphorous+	mg/kg	< 1	-	VLS-LM8
Zinc+	mg/kg	< 1	-	VLS-LM8
Potassium	mg/kg	< 1	-	VLS-LM8
Lead	mg/kg	< 1	-	VLS-LM8
Magnesium	mg/kg	< 1	-	VLS-LM8
API Gravity	-	29.05	-	-
Net Specific Energy	MJ/kg	42.480	-	Calculated
Gross Specific Energy	MJ/kg	44.950	-	Calculated
Visual Appearance		Clear & Bright	-	LM**

LM** - LAB METHOD

Calcium > 30, Zinc > 15 and Phosphorus > 15 (ISO 8217:2005 standard)

VESSEL PAST BUNKERING HISTORY DATA					
		SS261083440			
BUNKER DATE		21-Oct-2026			
BUNKER PORT		UST – LUGA			
QUANTITY (MT)		410.000			
TEST DESCRIPTION	UNITS				
Density @15°C	kg/m³	828.5			
Viscosity @ 40°C	cSt	2.69			
Flash Point	°C	> 70.0			
Pour Point	°C	0			
Micro Carbon Residue	% m/m	0.01			
Ash	% m/m	0.003			
Water	% v/v	< 0.01			
Sulfur	% m/m	0.00056			
Vanadium	mg/kg	< 1			
Al+Si (Catfines)	mg/kg	2			
Sodium	mg/kg	< 1			
Aluminum	mg/kg	< 1			
Silicon	mg/kg	< 1			
Iron	mg/kg	< 1			
Nickel	mg/kg	< 1			
Calcium	mg/kg	< 1			
Phosphorous	mg/kg	< 1			
Zinc	mg/kg	< 1			
Potassium	mg/kg	< 1			
Lead	mg/kg	< 1			
Magnesium	mg/kg	< 1			
API Gravity		29.05			
Net Specific Energy	MJ/kg	42.480			
Gross Specific Energy	MJ/kg	44.950			
Visual Appearance		Clear & Bright			

On behalf of Viswa Lab

Signed:

VIGNESH KUMAR S

Date: 15-Nov-2026

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- *Lab Name and Address : Viswa Lab Singapore, 80 Tagore Lane, Singapore 787501*

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